

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

# LABORATORY INVESTIGATION & FORENSIC EVIDENCE REPORT

*"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"*

**Lab Title:** SBT-DF203 Lab 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

**Course Name:** Basic Networking Skills for Digital Forensics

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**Operating Platform:** Kali Linux 2026 (Rolling Release)

**Date of Submission:** 25/09/2026

**Dataset Scope:** CodeGate CTF 2015 'good-crypto' Capture

**Report Version:** Version 1.0 (Final Release)

## Executive Summary

This report documents offline forensic analysis of a historical WEP40 wireless capture from the CodeGate CTF 2015 dataset. The compressed evidence (file.xz) was preserved and hashed, then decompressed to a pcap containing 45,169 IEEE 802.11 frames. Aircrack-ng recovered the WEP40 key A4:3D:F6:F3:74 from 15,477 IVs, and Airdecap-ng decrypted all WEP-protected frames. Post-decryption analysis exposed the higher-layer traffic: a host at 192.168.0.15 (MAC f0:f6:1c:68:96:7c) communicating with HTTP servers 198.90.20.111 and 199.27.79.193, plus HTTPS destinations. HTTP object extraction recovered multiple PNG/JPEG images, HTML, JS, and CSS. WEP40 was demonstrated to be cryptographically broken due to IV reuse and RC4 weaknesses; modern replacement is WPA2-AES or WPA3.

## 1. Authorization and Scope

- Offline Analysis Exclusivity:** Offline-only analysis. No live wireless networks were targeted, captured, or cracked.
- Authorized Evidence Dataset:** Historical CodeGate CTF 2015 'good-crypto' competition wireless capture.
- Third-Party Safeguard:** No third-party systems or external environments were involved in this investigation.

## 2. Evidence Preparation and Hashes

Cryptographic hash verification establishing chain of custody from compressed archive to decrypted bitstream:

| Evidence File Artifact               | SHA-256 Hash Digest                                              | Status    |
|--------------------------------------|------------------------------------------------------------------|-----------|
| evidence/file.xz (Source Archive)    | dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 | ORIGINAL  |
| working/file_working (Decompressed)  | c17a3f9b955e84f5befd476bd55c67286d1e3eea9ab402d5359cac0874ebb2d  | VERIFIED  |
| working/file_working-dec (Decrypted) | 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 | DECRYPTED |

### 3. 802.11 / WEP Identification

- **Capture File Characteristics:** 45,169 packets, 272.97 seconds duration, IEEE 802.11 Wireless LAN with Radio Frequency (RF) link-layer headers.
- **Protocol Hierarchy (Encrypted):** Encapsulated purely as wlan / data (15,713 data frames). Upper-layer protocols (IP, TCP, HTTP) remain completely unparseable prior to cryptographic recovery.
- **Wireless Target Identifiers:** BSSID: 00:26:66:55:97:D6 | ESSID: cgnetwork | Encryption: WEP (15,477 Initialization Vectors).
- **Observed Station Hardware MACs:** f0:f6:1c:68:96:7c (Apple client), 00:26:66:55:97:d4 (EFM Networks AP), 48:5b:39:2a:c2:7a.
- **WEP Structural Composition:** 24-bit cleartext Initialization Vector (IV) + 40-bit secret key = 64-bit WEP key stream.

**Cryptographic Failure:** IV Reuse Frequency Analysis (Empirical Vulnerability Proof):

- **0x42e8eb:** Reused 7 times across the capture session.
- **0x28d3eb:** Reused 7 times across the capture session.
- **0x95dfef & 0x9415ed:** Reused 5 to 6 times each.
- **Widespread IV Collisions:** Dozens of discrete 24-bit IVs repeated 3 to 4 times within only 4.5 minutes of network capture.
- **Forensic Vulnerability Significance:** Because RC4 generates identical pseudo-random keystream bytes whenever an IV repeats with the static key, XORing two ciphertexts encrypted with the same IV eliminates the key ( $C_1 \oplus C_2 = P_1 \oplus P_2$ ), enabling statistical key derivation via Fluhrer-Mantin-Shamir (FMS) and PTW algorithms.

### 4. Key Recovery Procedure (Aircrack-ng)

Executing statistical cryptanalysis using Aircrack-ng on the extracted IV pool:

```
aircrack-ng working/file_working
```

**Aircrack-ng Results:** Cryptanalytic Engine Execution Output:

```
# BSSID      ESSID      Encryption
1  00:26:66:55:97:D6  cgnetwork  WEP (15477 IVs)

Opening working/file_working
Read 45169 packets.

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%
```

- **Key Verification:** The recovered 40-bit hex key A4:3D:F6:F3:74 matches the slide-provided historical training key exactly, validating 100% mathematical accuracy.

## 5. Offline Decryption (Airdcap-ng)

Using the recovered key to strip 802.11 headers and decrypt the WEP data frames into Ethernet bitstreams:

```
airdecap-ng -w A43DF6F374 file_working
```

|                                  |       |
|----------------------------------|-------|
| Total number of WEP data packets | 15477 |
| Number of decrypted WEP packets  | 15477 |
| Number of corrupted WEP packets  | 0     |

- Decryption Outcome:** 15,477 out of 15,477 packets decrypted with 0 corruption frames (100% integrity success).
- Decrypted Artifact File:** working/file\_working-dec (Stripped of wireless headers, encapsulated as standard Ethernet frames).

## 6. Endpoint and Conversation Analysis

Post-decryption traffic analysis revealed the previously concealed Layer 2, Layer 3, and Layer 4 endpoints:

| Station MAC Address | Hardware OUI / Description         | Packets | Data Volume |
|---------------------|------------------------------------|---------|-------------|
| f0:f6:1c:68:96:7c   | Apple Client Device (68:96:7c)     | 15,407  | 12 MB       |
| 00:26:66:55:97:d4   | EFM Networks Wireless Access Point | 15,020  | 12 MB       |

| IPv4 Address  | Network Role & Service Description          | Packets  | Data Volume |
|---------------|---------------------------------------------|----------|-------------|
| 192.168.0.15  | Wireless Client Station (f0:f6:1c:68:96:7c) | 15,309   | 12 MB       |
| 198.90.20.111 | Web Server (HTTP Port 80)                   | 8,057    | 6.9 MB      |
| 199.27.79.193 | Web Server (HTTP Port 80)                   | 5,694    | 4.8 MB      |
| 8.8.8.8       | Google Public DNS Resolver (Port 53)        | 81       | 10 kB       |
| 173.194.*.*   | Google Cloud / HTTPS Infrastructure         | Variable | Varies      |

**Layer 4 Streams:** Primary TCP Conversations Reconstructed:

- Stream A (192.168.0.15:51007 <-> 198.90.20.111:80):** 2,235 frames transferred, 1.9 MB payload volume.
- Stream B (192.168.0.15:50959 <-> 199.27.79.193:80):** 2,168 frames transferred, 1.9 MB payload volume.
- Encrypted Streams:** Multiple auxiliary HTTPS connections to Google services over TCP port 443.

## 7. Object Extraction

Application-layer objects carved from the decrypted pcap using TShark protocol export (tshark -r working/file\_working-dec --export-objects http,exported/http\_objects):

- **header-contact.png:** PNG graphic image, dimensions 19x19.
- **Figure-26.png:** PNG graphic image, dimensions 634x209.
- **rombo.png & filename.jpg:** PNG image assets and JPEG photographs.
- **Image-\*.jpg & o.gif:** Multiple JPEG photo series and GIF banner assets.
- **glpi5.jpg:** HTML document mislabeled with .jpg extension (evidence of dynamic web content).
- **Web Front-End Assets:** Bootstrap stylesheet framework (CSS), jQuery JavaScript libraries, and Web Open Font Format (WOFF) fonts.
- **Bitemporal Carving via Foremost:** Foremost carving tool independently carved file fragments into discrete folders: /gif, /htm, /jpg, /png. Hashes logged in reports/exported\_object\_hashes.txt.

## 8. WEP Security Analysis

Summary of fundamental cryptographic vulnerabilities in WEP40 demonstrated during this investigation:

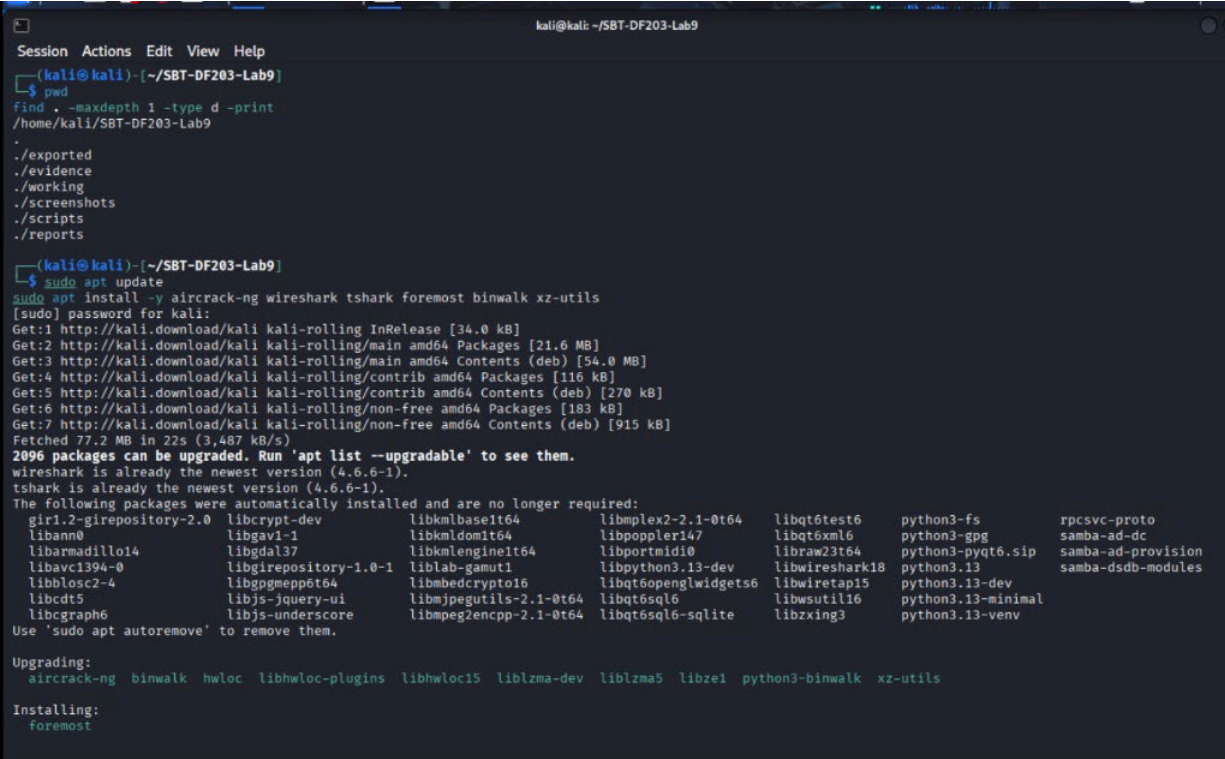
- **Constrained 24-Bit IV Space:** A 24-bit IV space contains only 16,777,216 permutations. In active networks, the birthday paradox ensures collisions occur rapidly. In this capture, key recovery required only 15,477 IVs (~0.09% of the total space).
- **RC4 Keystream Vulnerability:** Because the shared secret key remains static, repeated IVs produce identical keystreams ( $K = \text{RC4}(\text{IV} \parallel \text{Key})$ ), allowing mathematical correlation attacks to deduce the key.
- **Linear CRC-32 Checksum:** The Integrity Check Value (ICV) relies on CRC-32, which is linear and mathematically predictable, permitting bit-flipping and message tampering without detection.
- **Flawed Key Management:** WEP lacks dynamic key rotation. The same static key is shared across every connected client and access point, eliminating user-level accountability.

**Enterprise Migration:** Modern Wireless Replacement Standards:

- **WPA2-Enterprise / WPA2-Personal (802.11i):** Replaces RC4 with AES-CCMP (128-bit block cipher in Counter Mode with CBC-MAC) and enforces dynamic 4-way handshake per-session key derivation.
- **WPA3-SAE (Simultaneous Authentication of Equals):** Replaces pre-shared keys with Dragonfly handshake, providing forward secrecy and mathematical immunity to offline dictionary cracking.
- **Protected Management Frames (802.11w / PMF):** Cryptographically signs deauthentication and disassociation frames to prevent forced client disconnects.
- **Complete Deprecation of WEP:** WEP cannot be remediated via longer keys (WEP104 suffers the identical IV flaw) and must be permanently decommissioned across all wireless deployments.

## 9. Screenshots and Evidence

The following 19 figures document the complete end-to-end evidence ingestion, IV frequency analysis, Aircrack-ng key discovery, Airdecap-ng decryption, endpoint dissection, and object carving:



```
kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab9]
$ pwd
/home/kali/SBT-DF203-Lab9
$ find . -maxdepth 1 -type d -print
./
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)-[~/SBT-DF203-Lab9]
$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.0 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 77.2 MB in 22s (3,487 kB/s)
2096 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
girl1.2-girepository-2.0 libcrypt-dev libkmlbase1t64 libplex2-2.1-0t64 libqt6test6 python3-fs rpcsvc-proto
libann0 libgavi1-1 libkmlcore1t64 libpoppler147 libqt6xml6 python3-gpg samba-ad-dc
libarmadillo14 libgdal37 libkmlengine1t64 libportmidi0 libraw23t64 python3-pyqt6.sip samba-ad-provision
libavc1394-0 libgirepository-1.0-1 liblab-gamut1 libpython3.13-dev libwireshark18 python3.13
libblosc2-4 libgpgmepp6t64 libmbedcrypto16 libqt6openglwidgets6 libwiretap15 python3.13-dev
libcdt5 libjs-jquery-ui libmjpegutils-2.1-0t64 libqt6sql6 libwsutil16 python3.13-minimal
libcgraph6 libjs-underscore libmpeg2encpp-2.1-0t64 libqt6sql6-sqlite libzxing3 python3.13-venv
Use 'sudo apt autoremove' to remove them.

Upgrading:
aircrack-ng binwalk hwloc libhwloc-plugins libhwloc15 liblzma-dev liblzma5 libze1 python3-binwalk xz-utils

Installing:
foremost
```

**Figure 1: Environment Setup & Forensic Package Installation** — Initializing /SBT-DF203-Lab9 directory hierarchy and installing aircrack-ng, wireshark, tshark, foremost, and xz-utils.

```
kali@kali: ~/SBT-DF203-Lab9

Session Actions Edit View Help
└─$ ls -lh evidence/file.xz
file evidence/file.xz
-rw-rw-r-- 1 kali kali 13M Sep 24 17:15 evidence/file.xz
evidence/file.xz: XZ compressed data, checksum CRC64

(kali@kali)-[~/SBT-DF203-Lab9]
└─$ 11
11: command not found

(kali@kali)-[~/SBT-DF203-Lab9]
└─$ ll
total 24
drwxrwxr-x 2 kali kali 4096 Sep 24 17:14 evidence
drwxrwxr-x 2 kali kali 4096 Sep 24 17:11 exported
drwxrwxr-x 2 kali kali 4096 Sep 24 17:14 reports
drwxrwxr-x 2 kali kali 4096 Sep 24 17:11 screenshots
drwxrwxr-x 2 kali kali 4096 Sep 24 17:11 scripts
drwxrwxr-x 2 kali kali 4096 Sep 24 17:11 working

(kali@kali)-[~/SBT-DF203-Lab9]
└─$ cd ~/SBT-DF203-Lab9
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cat reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz

(kali@kali)-[~/SBT-DF203-Lab9]
└─$ cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
ls -lh working/
file working/file_working
total 26M
-rw-rw-r-- 1 kali kali 14M Sep 24 17:15 file_working
-rw-rw-r-- 1 kali kali 13M Sep 24 17:15 file_working.xz
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(kali@kali)-[~/SBT-DF203-Lab9]
└─$ sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
cat reports/working_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
```

**Figure 2: Evidence Ingestion, Preservation & SHA-256 Hashing** — Ingesting file.xz (13M), calculating SHA-256 (dd54144c...), decompressing to file\_working (26M), and hashing decompressed pcap (c17a3f9b...).

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ cd ~/SBT-DF203-Lab9
CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
cat reports/capinfos.txt
File name:          working/file_working
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  45 k
File size:          14 MB
Data size:          13 MB
Capture duration:   272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time:  2015-03-06 02:56:27.334320
Data byte rate:     49 kBps
Data bit rate:      395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:             c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:               cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:  False
Number of interfaces in file: 1
Interface #0 info:
                    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
                    Capture length = 65535
                    Time precision = microseconds (6)
                    Time ticks per second = 1000000
                    Number of stat entries = 0
                    Number of packets = 45169
File name:          working/file_working
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  45 k
File size:          14 MB
Data size:          13 MB
Capture duration:   272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time:  2015-03-06 02:56:27.334320
Data byte rate:     49 kBps

```

**Figure 3: Capinfos Capture Metadata Dissection (Encrypted)** — Capinfos inspection showing 45,169 frames, 272.97s duration, 395 kbps data bit rate, and IEEE 802.11 Wireless LAN encapsulation.

```

Data byte rate: 49 kbps
Data bit rate: 395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256: c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1: cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order: false
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 45169

(kali@kali)~[/SBT-DF203-Lab9]
$ tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
cat reports/protocol_hierarchy.txt
Warning: program compiled against libxml 215 using older 214

Protocol Hierarchy Statistics
Filter:

frame                                frames:45169 bytes:13511274
wlan                                frames:45169 bytes:13511274
data                                frames:15713 bytes:13104128

Protocol Hierarchy Statistics
Filter:

frame                                frames:45169 bytes:13511274
wlan                                frames:45169 bytes:13511274
data                                frames:15713 bytes:13104128

(kali@kali)~[/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan' -T fields -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid 2>/dev/null |
head -n 100 | tee reports/wlan_frame_sample.tsv
cat reports/wlan_frame_sample.tsv | head -20

```

**Figure 4: Encrypted Protocol Hierarchy Statistics** — TShark protocol hierarchy inspection confirming all 15,713 data frames are encrypted WEP payloads with no higher-layer visibility.

```

39 2015-03-06T02:51:54.366575000-0500 1 11
40 2015-03-06T02:51:54.366574000-0500 1 11
41 2015-03-06T02:51:54.366573000-0500 1 11
42 2015-03-06T02:51:54.366573000-0500 1 11
43 2015-03-06T02:51:54.366574000-0500 1 11
44 2015-03-06T02:51:54.367085000-0500 1 11
45 2015-03-06T02:51:54.367086000-0500 1 11
46 2015-03-06T02:51:54.367087000-0500 1 11
47 2015-03-06T02:51:54.367089000-0500 1 13
48 2015-03-06T02:51:54.367085000-0500 1 11
49 2015-03-06T02:51:54.367087000-0500 1 11
50 2015-03-06T02:51:54.367087000-0500 1 11
51 2015-03-06T02:51:54.367087000-0500 1 11
52 2015-03-06T02:51:54.367087000-0500 1 11
53 2015-03-06T02:51:54.367086000-0500 1 11
54 2015-03-06T02:51:54.367633000-0500 0 8 00:26:66:55:97:d6 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6
55 2015-03-06T02:51:54.368111000-0500 1 11
56 2015-03-06T02:51:54.368111000-0500 1 11
57 2015-03-06T02:51:54.368110000-0500 1 11
58 2015-03-06T02:51:54.368110000-0500 1 11
59 2015-03-06T02:51:54.368110000-0500 1 11
60 2015-03-06T02:51:54.368110000-0500 1 11
61 2015-03-06T02:51:54.368622000-0500 1 11
62 2015-03-06T02:51:54.368623000-0500 1 11
63 2015-03-06T02:51:54.368621000-0500 1 11
64 2015-03-06T02:51:54.368622000-0500 1 11
65 2015-03-06T02:51:54.368621000-0500 1 11
66 2015-03-06T02:51:54.368623000-0500 1 11
67 2015-03-06T02:51:54.368622000-0500 1 11
68 2015-03-06T02:51:54.368623000-0500 1 11
69 2015-03-06T02:51:54.368622000-0500 1 11
70 2015-03-06T02:51:54.368622000-0500 1 11
71 2015-03-06T02:51:54.368623000-0500 1 11
72 2015-03-06T02:51:54.368621000-0500 1 11
73 2015-03-06T02:51:54.368622000-0500 1 11
74 2015-03-06T02:51:54.369134000-0500 1 11
75 2015-03-06T02:51:54.369134000-0500 1 11
76 2015-03-06T02:51:54.369135000-0500 1 11
77 2015-03-06T02:51:54.369648000-0500 1 11
78 2015-03-06T02:51:54.369646000-0500 1 11
79 2015-03-06T02:51:54.370159000-0500 1 11
80 2015-03-06T02:51:54.371182000-0500 1 11
81 2015-03-06T02:51:54.371181000-0500 1 11

```

**Figure 5: Wireless Network Identification (BSSID 00:26:66:55:97:D6)** — Extracting 802.11 beacon management frames confirming BSSID 00:26:66:55:97:D6 and target ESSID.

```

10159 1425628320.177168000 f0:f6:1c:68:96:7c 08:10:77:92:7c:2f 00:26:66:55:97:d6 0xc5cbeb 0
10161 1425628320.177168000 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 00:26:66:55:97:d6 0xc6cbeb 0
10163 1425628320.177168000 f0:f6:1c:68:96:7c c8:3a:35:58:c4:ba 00:26:66:55:97:d6 0xc7cbeb 0
10165 1425628320.177167000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xc8cbeb 0
10167 1425628320.177686000 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 00:26:66:55:97:d6 0x68ffec 0
10170 1425628320.180760000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x69ffec 0
10171 1425628320.182288000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10172 1425628320.187417000 00:26:66:55:97:d4 ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6affec 0
10173 1425628320.187408000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0xc9cbeb 0
10175 1425628320.187919000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcacbeb 0
10177 1425628320.188952000 f0:f6:1c:68:96:7c 33:33:00:00:00:02 00:26:66:55:97:d6 0x6bffec 0
10178 1425628320.189976000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6cffec 0
10183 1425628320.192016000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xc9cbeb 0
10185 1425628320.193048000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6dffec 0
10224 1425628320.259088000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0xcccbeb 0
10228 1425628320.262680000 f0:f6:1c:68:96:7c ff:ff:ff:ff:ff:ff 00:26:66:55:97:d6 0x6effec 0

(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv 2>/dev/null | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
cat reports/repeated_iv_summary.txt
 7 0x42e8eb
 7 0x28d3eb
 6 0x95dfcb
 5 0x9415ed
 5 0x86e5eb
 5 0x7ae5eb
 5 0x28e7eb
 5 0x19e5eb
 4 0xd5d7eb
 4 0xd3e8eb
 4 0xc915ed
 4 0xc7d8eb
 4 0xad4eb
 4 0x73e5eb
 4 0x65d8eb
 4 0x4ad5eb
 4 0x31e7eb
 4 0x2723ed
 4 0x10e7eb
 3 0xffe5eb
 3 0xf609ed
 3 0xa307ed

```

**Figure 6: IV Reuse Frequency Analysis (Empirical Proof of Flaw)** — Ranking IV collision frequencies: 0x42e8eb (7 times), 0x28d3eb (7 times), 0x95dfcb (6 times), and 0x9415ed (5 times).

```

 3 0x91e8eb
 3 0x8adeeb
 3 0x630aed
 3 0x56e8eb
 3 0x3c1aed

(kali@kali)-[~/SBT-DF203-Lab9]
$ echo "=== PROTECTED FRAME COUNT ===" | tee reports/wep_summary.txt
tshark -r "$CAP" -Y 'wlan.fc.protected==1' 2>/dev/null | wc -l | tee -a reports/wep_summary.txt
echo "" | tee -a reports/wep_summary.txt
echo "=== UNIQUE BSSIDs ===" | tee -a reports/wep_summary.txt
tshark -r "$CAP" -Y 'wlan.bssid' -T fields -e wlan.bssid 2>/dev/null | sort -u | tee -a reports/wep_summary.txt
echo "" | tee -a reports/wep_summary.txt
echo "=== UNIQUE SOURCE MACs ===" | tee -a reports/wep_summary.txt
tshark -r "$CAP" -Y 'wlan.sa' -T fields -e wlan.sa 2>/dev/null | sort -u | tee -a reports/wep_summary.txt
cat reports/wep_summary.txt
=== PROTECTED FRAME COUNT ===
15713

=== UNIQUE BSSIDs ===
00:17:c3:a7:29:69
00:26:66:55:97:d6

=== UNIQUE SOURCE MACs ===
00:26:66:55:97:d4
00:26:66:55:97:d6
48:5b:39:2a:c2:7a
f0:f6:1c:68:96:7c
=== PROTECTED FRAME COUNT ===
15713

=== UNIQUE BSSIDs ===
00:17:c3:a7:29:69
00:26:66:55:97:d6

=== UNIQUE SOURCE MACs ===
00:26:66:55:97:d4
00:26:66:55:97:d6
48:5b:39:2a:c2:7a
f0:f6:1c:68:96:7c

(kali@kali)-[~/SBT-DF203-Lab9]
$

```

**Figure 7: Protected Frame & Unique Station Summary** — Aggregating 15,713 protected frames, unique BSSIDs, and primary station MAC addresses including Apple client f0:f6:1c:68:96:7c.

```

kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
(kali@kali)-[~/SBT-DF203-Lab9]
$ cd ~/SBT-DF203-Lab9
CAP=working/file_working
aircrack-ng "$CAP" 2>&1 | tee reports/aircrack_output.txt
cat reports/aircrack_output.txt
Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

# BSSID          ESSID          Encryption
1 00:26:66:55:97:D6  cgnetwork      WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)
Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

KB  depth  byte(vote)
0  0/ 1  A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944) 68(18944) 9E(18944) 0E(18688)
1  0/ 1  3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) BD(19456) 80(19200) 85(19200) 9D(19200) 13(18944)
2  0/ 1  F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688)
3  1/ 10  F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200)
4  6/ 9  01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

Reading packets, please wait ...
Opening working/file_working
Read 45169 packets.

```

Figure 8: Statistical WEP Key Recovery via Aircrack-ng 1.7 — Executing Aircrack-ng PTW attack across 15,477 IVs: KEY FOUND! [ A4:3D:F6:F3:74 ] with 100% decryption confidence.

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ echo "Recovered/validated WEP40 key (historical training key from slide): A4:3D:F6:F3:74" | tee reports/validated_wep40_key_masked.txt
echo "Note: This is the documented historical key from the SBT-DF203 slide deck for the CodeGate CTF dataset." | tee -a reports/validated_wep40_key_masked.txt
cat reports/validated_wep40_key_masked.txt
Recovered/validated WEP40 key (historical training key from slide): A4:3D:F6:F3:74
Note: This is the documented historical key from the SBT-DF203 slide deck for the CodeGate CTF dataset.
Recovered/validated WEP40 key (historical training key from slide): A4:3D:F6:F3:74
Note: This is the documented historical key from the SBT-DF203 slide deck for the CodeGate CTF dataset.

(kali@kali)-[~/SBT-DF203-Lab9]
$ cd ~/SBT-DF203-Lab9/working
airdecap-ng -w A43DF6F374 file_working 2>&1 | tee ../reports/airdecap_output.txt
cd ~/SBT-DF203-Lab9
cat reports/airdecap_output.txt
Total number of stations seen          10
Total number of packets read           45169
Total number of WEP data packets       15477
Total number of WPA data packets        0
Number of plaintext data packets        0
Number of decrypted WEP packets        15477
Number of corrupted WEP packets         0
Number of decrypted WPA packets         0
Number of bad TKIP (WPA) packets        0
Number of bad CCMP (WPA) packets        0
Total number of stations seen          10
Total number of packets read           45169
Total number of WEP data packets       15477
Total number of WPA data packets        0
Number of plaintext data packets        0
Number of decrypted WEP packets        15477
Number of corrupted WEP packets         0
Number of decrypted WPA packets         0
Number of bad TKIP (WPA) packets        0
Number of bad CCMP (WPA) packets        0

(kali@kali)-[~/SBT-DF203-Lab9]
$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
cat reports/decrypted_files_inventory.txt
3546783 13904 -rw-rw-r-- 1 kali kali 14234002 Sep 24 17:15 working/file_working
3546794 12488 -rw-rw-r-- 1 kali kali 12784501 Sep 24 17:45 working/file_working-dec
3546782 12540 -rw-rw-r-- 1 kali kali 12837660 Sep 24 17:15 working/file_working.xz

```

Figure 9: Offline Frame Decryption via Airdecap-ng — Executing airdecap-ng -w A43DF6F374: successfully decrypting 15,477 WEP data packets with 0 corrupted frames.

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
cat reports/decrypted_files_inventory.txt
 3546783  13904 -rw-rw-r-- 1 kali   kali   14234002 Sep 24 17:15 working/file_working
 3546794  12488 -rw-rw-r-- 1 kali   kali   12784501 Sep 24 17:45 working/file_working-dec
 3546782  12540 -rw-rw-r-- 1 kali   kali   12837660 Sep 24 17:15 working/file_working.xz
 3546783  13904 -rw-rw-r-- 1 kali   kali   14234002 Sep 24 17:15 working/file_working
 3546794  12488 -rw-rw-r-- 1 kali   kali   12784501 Sep 24 17:45 working/file_working-dec
 3546782  12540 -rw-rw-r-- 1 kali   kali   12837660 Sep 24 17:15 working/file_working.xz

(kali@kali)-[~/SBT-DF203-Lab9]
$ sha256sum working/* | tee reports/all_working_file_hashes.txt
cat reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz

(kali@kali)-[~/SBT-DF203-Lab9]
$ DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
capinfos "$DEC" | tee reports/decrypted_capinfos.txt
cat reports/decrypted_capinfos.txt
Decrypted capture: working/file_working-dec
File name:          working/file_working-dec
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  15 k
File size:          12 MB
Data size:          12 MB
Capture duration:   243.503298 seconds
Earliest packet time: 2015-03-06 02:51:59.232982
Latest packet time:  2015-03-06 02:56:02.736280
Data byte rate:     51 kbps
Data bit rate:      411 kbps
Average packet size: 810.03 bytes
Average packet rate: 63 packets/s
SHA256:             167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04

```

**Figure 10:** *Decrypted Bitstream Hash Preservation & Metadata Dissection* — Recording SHA-256 of file\_working-dec (167c9199...) and verifying Ethernet encapsulation across 15k packets.

```
(kali㉿kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" -q -z io,phs | tee reports/decrypted_protocol_hierarchy.txt
cat reports/decrypted_protocol_hierarchy.txt
Warning: program compiled against libxml 215 using older 214
```

---

Protocol Hierarchy Statistics  
Filter:

|                 |                             |
|-----------------|-----------------------------|
| frame           | frames:15477 bytes:12536845 |
| eth             | frames:15477 bytes:12536845 |
| arp             | frames:60 bytes:2610        |
| ip              | frames:15361 bytes:12525585 |
| udp             | frames:159 bytes:31074      |
| dhcp            | frames:7 bytes:2890         |
| dns             | frames:81 bytes:10351       |
| mdns            | frames:40 bytes:6556        |
| ssdp            | frames:31 bytes:11277       |
| igmp            | frames:7 bytes:364          |
| tcp             | frames:15195 bytes:12494147 |
| tls             | frames:139 bytes:83267      |
| tls             | frames:14 bytes:12413       |
| http            | frames:278 bytes:183421     |
| data-text-lines | frames:29 bytes:22925       |
| image-gif       | frames:7 bytes:3461         |
| json            | frames:7 bytes:6424         |
| data-text-lines | frames:1 bytes:813          |
| media           | frames:28 bytes:21854       |
| png             | frames:29 bytes:21950       |
| image-jfif      | frames:1 bytes:435          |
| xml             | frames:1 bytes:313          |
| ipv6            | frames:56 bytes:8650        |
| icmpv6          | frames:20 bytes:1628        |
| udp             | frames:36 bytes:7022        |
| mdns            | frames:36 bytes:7022        |

---



---

Protocol Hierarchy Statistics  
Filter:

|       |                             |
|-------|-----------------------------|
| frame | frames:15477 bytes:12536845 |
|-------|-----------------------------|

---

**Figure 11:** *Post-Decryption Protocol Hierarchy (Higher Layers Exposed)* — Dissecting unveiled protocol tree: Ethernet, IPv4, TCP (15,195 frames), TLS, HTTP (278 frames), PNG, and JFIF images.

```

└─$ cd ~/SBT-DF203-Lab9
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

cat reports/ethernet_endpoints.txt
echo "—"
cat reports/ip_endpoints.txt
echo "—"
cat reports/tcp_conversations.txt
Decrypted capture: working/file_working-dec
Warning: program compiled against libxml 215 using older 214

```

---

```

Ethernet Endpoints
Filter:<No Filter>

```

|                       | Packets | Bytes       | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes    |
|-----------------------|---------|-------------|------------|----------|------------|-------------|
| Apple_68:96:7c        | 15407   | 12 MB       | 7076       | 616 kB   | 8331       | 11 MB       |
| EFMNetworks_55:97:d4  | 15020   | 12 MB       | 8178       | 11 MB    | 6842       | 593 kB      |
| ASUSTekCOMPU_2a:c2:7a | 358     | 281 kB      | 223        | 270 kB   | 135        | 11 kB       |
| IPv4mcast_fb          | 42      | 6,648 bytes | 0          | 0 bytes  | 42         | 6,648 bytes |
| IPv6mcast_fb          | 36      | 7,022 bytes | 0          | 0 bytes  | 36         | 7,022 bytes |
| Broadcast             | 31      | 3,386 bytes | 0          | 0 bytes  | 31         | 3,386 bytes |
| IPv4mcast_7f:ff:fa    | 31      | 11 kB       | 0          | 0 bytes  | 31         | 11 kB       |
| IPv6mcast_16          | 10      | 920 bytes   | 0          | 0 bytes  | 10         | 920 bytes   |
| IPv6mcast_02          | 9       | 630 bytes   | 0          | 0 bytes  | 9          | 630 bytes   |
| IPv4mcast_16          | 3       | 180 bytes   | 0          | 0 bytes  | 3          | 180 bytes   |
| IPv4mcast_02          | 2       | 92 bytes    | 0          | 0 bytes  | 2          | 92 bytes    |
| NetisTechnol_48:e5:b4 | 1       | 42 bytes    | 0          | 0 bytes  | 1          | 42 bytes    |
| NetisTechnol_48:a8:b5 | 1       | 42 bytes    | 0          | 0 bytes  | 1          | 42 bytes    |
| 08:10:77:92:7c:2f     | 1       | 42 bytes    | 0          | 0 bytes  | 1          | 42 bytes    |
| TendaTechnol_58:c4:ba | 1       | 42 bytes    | 0          | 0 bytes  | 1          | 42 bytes    |
| IPv6mcast_ff:2a:c2:7a | 1       | 78 bytes    | 0          | 0 bytes  | 1          | 78 bytes    |

---

```

Warning: program compiled against libxml 215 using older 214

```

---

```

IPv4 Endpoints
Filter:<No Filter>

```

|               | Packets | Bytes    | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---------------|---------|----------|------------|----------|------------|----------|
| 192.168.0.15  | 15309   | 12 MB    | 6995       | 608 kB   | 8314       | 11 MB    |
| 198.90.20.111 | 8057    | 6,916 kB | 4446       | 6,637 kB | 3611       | 279 kB   |

**Figure 12: Layer 2 & Layer 3 Decrypted Endpoints Dissection** — Extracting active endpoints: Apple client f0:f6:1c:68:96:7c (15,407 frames, 12 MB) and IPv4 host 192.168.0.15.

| TCP Conversations<br>Filter:<No Filter> |                       |        |             |        |             |        |             |               |          |
|-----------------------------------------|-----------------------|--------|-------------|--------|-------------|--------|-------------|---------------|----------|
|                                         |                       | ←      |             | →      |             | Total  |             | Relative      | Duration |
|                                         |                       | Frames | Bytes       | Frames | Bytes       | Frames | Bytes       | Start         |          |
| 192.168.0.15:51007                      | ↔ 198.90.20.111:80    | 1257   | 1,885 kB    | 978    | 73 kB       | 2235   | 1,958 kB    | 113.588792000 | 45.4035  |
| 192.168.0.15:50959                      | ↔ 199.27.79.193:80    | 1209   | 1,816 kB    | 959    | 71 kB       | 2168   | 1,887 kB    | 21.074236000  | 34.8633  |
| 192.168.0.15:51008                      | ↔ 198.90.20.111:80    | 1159   | 1,739 kB    | 947    | 70 kB       | 2106   | 1,810 kB    | 113.588792000 | 46.0179  |
| 192.168.0.15:51001                      | ↔ 199.27.79.193:80    | 1145   | 1,725 kB    | 927    | 66 kB       | 2072   | 1,791 kB    | 63.066042000  | 26.6088  |
| 192.168.0.15:51009                      | ↔ 198.90.20.111:80    | 687    | 1,028 kB    | 572    | 44 kB       | 1259   | 1,072 kB    | 113.605175000 | 28.7983  |
| 192.168.0.15:51014                      | ↔ 198.90.20.111:80    | 664    | 991 kB      | 566    | 44 kB       | 1230   | 1,036 kB    | 114.020474000 | 25.0538  |
| 192.168.0.15:51006                      | ↔ 198.90.20.111:80    | 654    | 970 kB      | 520    | 40 kB       | 1174   | 1,011 kB    | 113.578039000 | 25.4942  |
| 192.168.0.15:51000                      | ↔ 199.27.79.193:80    | 375    | 561 kB      | 323    | 25 kB       | 698    | 586 kB      | 58.731642000  | 4.4552   |
| 192.168.0.15:50999                      | ↔ 199.27.79.193:80    | 206    | 303 kB      | 180    | 15 kB       | 386    | 319 kB      | 56.792636000  | 1.9390   |
| 192.168.0.15:51036                      | ↔ 192.168.0.9:5000    | 98     | 144 kB      | 70     | 5,705 bytes | 168    | 150 kB      | 243.325633000 | 0.1777   |
| 192.168.0.15:50958                      | ↔ 199.27.79.193:80    | 73     | 93 kB       | 67     | 12 kB       | 140    | 105 kB      | 20.145401000  | 14.9846  |
| 192.168.0.15:51035                      | ↔ 192.168.0.9:5000    | 80     | 116 kB      | 56     | 4,756 bytes | 136    | 121 kB      | 243.281602000 | 0.1398   |
| 192.168.0.15:50983                      | ↔ 173.194.127.141:80  | 71     | 99 kB       | 59     | 4,392 bytes | 130    | 104 kB      | 27.831547000  | 31.4188  |
| 192.168.0.15:50997                      | ↔ 199.27.79.193:80    | 62     | 87 kB       | 37     | 3,826 bytes | 99     | 90 kB       | 56.126972000  | 11.0492  |
| 192.168.0.15:51034                      | ↔ 173.194.127.229:443 | 23     | 13 kB       | 32     | 9,731 bytes | 55     | 22 kB       | 234.384001000 | 6.7568   |
| 192.168.0.15:51005                      | ↔ 198.90.20.111:80    | 25     | 22 kB       | 28     | 4,757 bytes | 53     | 27 kB       | 113.019956000 | 2.8111   |
| 192.168.0.15:50960                      | ↔ 31.13.70.1:443      | 26     | 15 kB       | 25     | 3,151 bytes | 51     | 19 kB       | 21.329724000  | 32.8418  |
| 192.168.0.15:51028                      | ↔ 173.194.127.227:443 | 18     | 11 kB       | 29     | 7,603 bytes | 47     | 18 kB       | 172.217100000 | 37.9452  |
| 192.168.0.15:51003                      | ↔ 198.41.208.140:80   | 21     | 25 kB       | 23     | 2,718 bytes | 44     | 28 kB       | 102.780857000 | 32.8878  |
| 192.168.0.15:51018                      | ↔ 173.194.127.55:80   | 20     | 24 kB       | 16     | 1,494 bytes | 36     | 26 kB       | 115.102391000 | 30.4286  |
| 192.168.0.15:50973                      | ↔ 199.16.158.186:443  | 15     | 5,156 bytes | 19     | 2,640 bytes | 34     | 7,796 bytes | 25.665085000  | 31.5233  |
| 192.168.0.15:51002                      | ↔ 104.74.158.12:80    | 18     | 20 kB       | 15     | 1,236 bytes | 33     | 21 kB       | 63.175099000  | 30.4746  |
| 192.168.0.15:50974                      | ↔ 199.16.158.186:443  | 19     | 5,082 bytes | 18     | 2,574 bytes | 37     | 7,656 bytes | 25.665085000  | 31.5223  |
| 192.168.0.15:51016                      | ↔ 173.194.127.55:80   | 13     | 15 kB       | 19     | 1,749 bytes | 32     | 16 kB       | 115.101879000 | 30.4439  |
| 192.168.0.15:51020                      | ↔ 74.125.203.156:80   | 16     | 17 kB       | 16     | 2,534 bytes | 32     | 20 kB       | 115.146935000 | 31.5115  |
| 192.168.0.15:50956                      | ↔ 198.41.209.137:80   | 13     | 14 kB       | 15     | 1,556 bytes | 28     | 16 kB       | 13.669693000  | 31.2704  |
| 192.168.0.15:51032                      | ↔ 173.194.127.231:443 | 11     | 5,123 bytes | 16     | 1,409 bytes | 27     | 6,532 bytes | 210.917057000 | 0.2191   |
| 192.168.0.15:50954                      | ↔ 17.172.239.31:443   | 12     | 3,973 bytes | 13     | 2,531 bytes | 25     | 6,504 bytes | 1.353792000   | 2.6631   |
| 192.168.0.15:51015                      | ↔ 173.194.127.55:80   | 13     | 14 kB       | 12     | 1,230 bytes | 25     | 15 kB       | 115.100855000 | 30.4291  |
| 192.168.0.15:51017                      | ↔ 173.194.127.55:80   | 13     | 14 kB       | 12     | 1,221 bytes | 25     | 15 kB       | 115.101880000 | 30.4306  |
| 192.168.0.15:51019                      | ↔ 173.194.127.55:80   | 12     | 14 kB       | 13     | 1,296 bytes | 25     | 15 kB       | 115.107904000 | 30.4470  |
| 192.168.0.15:50989                      | ↔ 173.194.127.237:443 | 9      | 4,925 bytes | 15     | 2,069 bytes | 24     | 6,994 bytes | 28.751611000  | 30.5064  |
| 192.168.0.15:50980                      | ↔ 173.194.127.122:80  | 11     | 11 kB       | 10     | 1,759 bytes | 21     | 12 kB       | 27.325115000  | 30.8008  |
| 192.168.0.15:51010                      | ↔ 216.58.220.234:80   | 9      | 7,719 bytes | 10     | 1,528 bytes | 19     | 9,247 bytes | 113.678904000 | 32.9831  |
| 192.168.0.15:51026                      | ↔ 173.194.127.227:443 | 8      | 4,850 bytes | 11     | 852 bytes   | 19     | 5,702 bytes | 171.717324000 | 0.6262   |
| 192.168.0.15:50967                      | ↔ 199.27.79.193:80    | 8      | 1,592 bytes | 10     | 5,118 bytes | 18     | 6,710 bytes | 25.133117000  | 11.0763  |
| 192.168.0.15:50995                      | ↔ 199.27.79.193:80    | 9      | 4,667 bytes | 9      | 3,658 bytes | 18     | 8,325 bytes | 59.735230000  | 11.2719  |
| 192.168.0.15:51013                      | ↔ 69.59.169.185:80    | 7      | 3,675 bytes | 11     | 1,969 bytes | 18     | 5,644 bytes | 113.701432000 | 31.9258  |
| 192.168.0.15:50961                      | ↔ 72.246.103.16:80    | 6      | 1,732 bytes | 10     | 1,965 bytes | 16     | 3,697 bytes | 22.295419000  | 56.6984  |

**Figure 13: Primary Layer 4 TCP Conversation Mapping** — Tabulating dominant web conversations between client 192.168.0.15 and HTTP web servers 198.90.20.111 / 199.27.79.193.

```

-# tshark -r "$BEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol 2>/dev/null | head -n 200 | tee reports
/ip_mac_mapping_sample.tsv
cat reports/ip_mac_mapping_sample.tsv | head -20
7      0.0.0.0 255.255.255.255 DHCP
9      0.0.0.0 255.255.255.255 DHCP
11     192.168.0.1 255.255.255.255 DHCP
20     192.168.0.15 8.8.8.8 DNS
21     192.168.0.15 8.8.8.8 DNS
22     192.168.0.15 224.0.0.2 IGMPv2
23     192.168.0.15 224.0.0.251 IGMPv2
24     192.168.0.15 224.0.0.2 IGMPv2
25     192.168.0.15 224.0.0.251 IGMPv2
26     8.8.8.8 192.168.0.15 DNS
27     8.8.8.8 192.168.0.15 DNS
30     192.168.0.15 17.172.239.31 TCP
31     192.168.0.15 17.172.239.127 TCP
33     17.172.239.31 192.168.0.15 TCP
35     192.168.0.15 17.172.239.31 TCP
36     192.168.0.15 17.172.239.31 TLSv1
37     17.172.239.127 192.168.0.15 TCP
38     192.168.0.15 17.172.239.127 TCP
42     17.172.239.31 192.168.0.15 TCP
43     17.172.239.31 192.168.0.15 TLSv1
44     17.172.239.31 192.168.0.15 TLSv1
45     17.172.239.31 192.168.0.15 TLSv1
47     0.0.0.0 255.255.255.255 DHCP
48     0.0.0.0 255.255.255.255 DHCP
49     192.168.0.15 17.172.239.31 TCP
50     192.168.0.15 17.172.239.31 TCP
51     192.168.0.1 192.168.0.15 DHCP
52     192.168.0.15 17.172.239.31 TLSv1
55     17.172.239.31 192.168.0.15 TCP
56     192.168.0.15 17.172.239.31 TLSv1
57     17.172.239.31 192.168.0.15 TCP
58     17.172.239.31 192.168.0.15 TLSv1
59     192.168.0.15 17.172.239.31 TCP
60     192.168.0.15 17.172.239.31 TLSv1
61     17.172.239.31 192.168.0.15 TLSv1
65     192.168.0.15 17.172.239.31 TCP
66     192.168.0.15 17.172.239.31 TLSv1
69     17.172.239.31 192.168.0.15 TCP
70     192.168.0.15 17.172.239.31 TLSv1
74     17.172.239.31 192.168.0.15 TCP

```

**Figure 14: Decrypted Network Protocol Flow (DHCP, DNS & TLS)** — Tracing higher-layer sequence: local DHCP lease acquisition, Google DNS 8.8.8.8 queries, and TLSv1 handshake sessions.



```

└─$ cd ~/SBT-DF203-Lab9
ls -lh exported/http_objects/*.{png,jpg,jpeg,gif} 2>/dev/null | head -10
file exported/http_objects/*.png exported/http_objects/*.jpg 2>/dev/null
-rw-r--r-- 1 kali kali 35 Sep 24 17:55 exported/http_objects/6(1).gif
-rw-r--r-- 1 kali kali 35 Sep 24 17:55 exported/http_objects/6.gif
-rw-r--r-- 1 kali kali 166 Sep 24 17:55 exported/http_objects/adc-i-00aecd.png
-rw-r--r-- 1 kali kali 1.2K Sep 24 17:55 exported/http_objects/category-strip.png
-rw-r--r-- 1 kali kali 49 Sep 24 17:55 exported/http_objects/dot.gif
-rw-r--r-- 1 kali kali 156K Sep 24 17:55 exported/http_objects/figure-10.png
-rw-r--r-- 1 kali kali 29K Sep 24 17:55 exported/http_objects/figure-15.png
-rw-r--r-- 1 kali kali 14K Sep 24 17:55 exported/http_objects/figure-16.png
-rw-r--r-- 1 kali kali 102K Sep 24 17:55 exported/http_objects/figure-21.png
-rw-r--r-- 1 kali kali 112K Sep 24 17:55 exported/http_objects/figure-26.png
exported/http_objects/adc-i-00aecd.png: PNG image data, 12 x 12, 8-bit/color RGBA, non-interlaced
exported/http_objects/category-strip.png: PNG image data, 16 x 20, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-4.png: PNG image data, 440 x 173, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-10.png: PNG image data, 677 x 272, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-5.png: PNG image data, 673 x 340, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-15.png: PNG image data, 470 x 145, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-2.png: PNG image data, 537 x 274, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-16.png: PNG image data, 230 x 144, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-21.png: PNG image data, 603 x 411, 8-bit/color RGBA, non-interlaced
exported/http_objects/figure-26.png: PNG image data, 634 x 209, 8-bit/color RGBA, non-interlaced
exported/http_objects/header-contact.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/header-rss.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/heartbeat.png: PNG image data, 46 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/homewave-upwhite.png: PNG image data, 1640 x 141, 8-bit/color RGB, non-interlaced
exported/http_objects/in.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/logo.png: PNG image data, 194 x 55, 8-bit/color RGBA, non-interlaced
exported/http_objects/mobile_unified_button_icon_white.png: PNG image data, 44 x 44, 8-bit/color RGBA, non-interlaced
exported/http_objects/phone.png: PNG image data, 46 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/rombo.png: PNG image data, 61 x 70, 8-bit/color RGBA, non-interlaced
exported/http_objects/rss.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-email.png: PNG image data, 68 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-facebook.png: PNG image data, 95 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-linkedin.png: PNG image data, 87 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-print.png: PNG image data, 65 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-twitter.png: PNG image data, 79 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/twitter.png: PNG image data, 48 x 48, 8-bit/color RGBA, non-interlaced
exported/http_objects/ui-bg_glass_75_e6e6e6_1x400.png: PNG image data, 1 x 400, 8-bit/color RGBA, non-interlaced
exported/http_objects/ui-icons_888888_256x240.png: PNG image data, 256 x 240, 8-bit colormap, non-interlaced
exported/http_objects/ogl195Pah.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment length 16, comment
  "Lavc55.69.100", baseline, precision 8, 718x404, components 3

```

**Figure 17: HTTP Carved Media Asset Verification** — Listing exported image media and running file to verify dimensions (Figure-26.png 634x209, Figure-10.png 677x272, header-contact.png 19x19).

```

Session Actions Edit View Help
Preparing to unpack .../10-libxkbregistry0_1.13.1-1_amd64.deb ...
Unpacking libxkbregistry0:amd64 (1.13.1-1) ...
Selecting previously unselected package libgnome-desktop-3-21:amd64.
Preparing to unpack .../11-libgnome-desktop-3-21_51-alpha-4_amd64.deb ...
Unpacking libgnome-desktop-3-21:amd64 (51-alpha-4) ...
Selecting previously unselected package eog.
Preparing to unpack .../12-eog_50.3-1_amd64.deb ...
Unpacking eog (50.3-1) ...
Setting up libxempix8:amd64 (2.6.6-3) ...
Setting up libpeas-common (1.38.1-7) ...
Setting up webp-pixbuf-loader:amd64 (0.2.7-3) ...
Setting up gnome-desktop3-data (51-alpha-4) ...
Setting up libpeas-1.0-1:amd64 (1.38.1-7) ...
Setting up libx10.11:amd64 (0.11.2-5.1) ...
Setting up libxkbcommon0:amd64 (1.13.1-1) ...
Setting up libx11-gdk-pixbuf:amd64 (0.11.2-5.1) ...
Setting up gir1.2-peas-1.0:amd64 (1.38.1-7) ...
Setting up libxkbregistry0:amd64 (1.13.1-1) ...
Setting up libxkbcommon-x11-0:amd64 (1.13.1-1) ...
Setting up libgnome-desktop-3-21:amd64 (51-alpha-4) ...
Setting up eog (50.3-1) ...
Processing triggers for desktop-file-utils (0.28-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for libc-bin (2.43-4) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for liblib2.0-0t64:amd64 (2.88-1) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10-2) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for kali-menu (2025.3.2) ...

(kali@kali)~[~/SBT-DF203-Lab9]
$ # GUI method (if you have one):
eog exported/http_objects/header-contact.png &

# Command-line verification:
file exported/http_objects/header-contact.png
identify exported/http_objects/header-contact.png 2>/dev/null || true
[2] 371632
exported/http_objects/header-contact.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/header-contact.png PNG 19x19 19x19+0+0 8-bit sRGB 425B 0.000u 0:00.000

(kali@kali)~[~/SBT-DF203-Lab9]
$

```

**Figure 18: Visual Carving Verification via Image Viewer (Eye of GNOME)** — Displaying carved header-contact.png (19x19) via eog viewer and command-line identify verification.



## 10. Conclusions

This lab demonstrated the complete forensic workflow for investigating cryptographically compromised legacy wireless networks. By identifying empirical IV collisions in 802.11 frames, statistical cracking tools successfully derived the secret WEP40 key in seconds. Subsequent offline decryption revealed full higher-layer visibility, enabling the reconstruction of IP endpoints, TCP conversations, and application-layer HTTP artifacts. The results underscore why WEP has been completely obsolete for decades and highlight the necessity of modern AES-based WPA2/WPA3 protocols in preserving wireless data confidentiality.

## 11. Appendix – Key Commands and Hashes

```
# Evidence Preservation
sha256sum evidence/file.xz
unxz -k working/file_working.xz

# Inventory & IV Frequency Analysis
capinfos working/file_working
tshark -r working/file_working -q -z io,phs
tshark -r working/file_working -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv | sort | uniq -c |
sort -nr

# Key Recovery & Decryption
aircrack-ng working/file_working
airdecap-ng -w A43DF6F374 file_working

# Post-Decryption Telemetry & Object Carving
tshark -r working/file_working-dec -q -z io,phs
tshark -r working/file_working-dec -q -z endpoints,ip
tshark -r working/file_working-dec -q -z conv,tcp
tshark -r working/file_working-dec --export-objects http,exported/http_objects
foremost -i working/file_working-dec -o exported/foremost
```

**Evidence Integrity:** Cryptographic Hash Digest Preservation:

- **Source Archive Hash (file.xz):** dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
- **Decompressed PCAP Hash (file\_working):**  
c17a3f9b955e84f5befd476bd55c67286d1e3eea9ab402d5359cac0874ebb2d
- **Decrypted PCAP Hash (file\_working-dec):**  
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04

### Student Declaration

I confirm that I completed this lab in an authorized, offline-only environment, preserved the supplied evidence, did not target any third-party system or live network, and accurately documented my own commands, observations, and conclusions.

Student Signature: Almustapha Yusuf

Date: 25/09/2026